

Ordinals and Typed Lambda Calculus

Numerical Systems in the Lambda Calculus

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Church Encoding

Booleans

We encoding the Booleans.

$\text{true} := \lambda x y. x$, where $\text{true } M N =_{\beta} M$.

$\text{false} := \lambda x y. y$, where $\text{false } M N =_{\beta} N$.

Church Encoding

Definition

For $n \in \mathbb{N}$ and $F, M \in \Delta$, we define

$$\begin{aligned} F^0 M &:= M, \\ F^{n+1} M &:= F (F^n M). \end{aligned}$$

Church Encoding

Definition

The **Church numerals**, denoted by c_n with $n \in \mathbb{N}$, encode the natural numbers.

- One definition

$$c_n := \lambda f x. f^n x,$$

- Other definition

$$c_0 := \lambda f x. x,$$

$$\text{succ} := \lambda n f x. f (n f x),$$

$$c_{n+1} := \text{succ } c_n.$$

Church Encoding

Example

β -nf's for some numerals.

$$c_0 =_{\beta} \lambda f x. x,$$

$$c_1 =_{\beta} \lambda f x. f x,$$

$$c_2 =_{\beta} \lambda f x. f (f x),$$

$$c_3 =_{\beta} \lambda f x. f (f (f x)).$$

Whiteboard.

Church Encoding

Addition, multiplication and exponentiation

Encoding for the arithmetic operations.

$\text{add} := \lambda m n f x. m f (n f x)$, where

$\text{add } c_m c_n =_{\beta} c_{m+n}$.

$\text{mult} := \lambda m n f x. m (n f) x$
 $\equiv \lambda m n f. m (n f)$, where

$\text{mult } c_m c_n =_{\beta} c_{m \times n}$.

$\text{exp} := \lambda m n f x. (n m) f x$, where

$\text{exp } c_m c_n =_{\beta} c_{m^n}$.

Church Encoding

Testing for zero

A combinator for testing for zero is defined by

$$\text{isZero} := \lambda n. n (\lambda x. \text{false}) \text{true},$$

where

$$\begin{aligned} \text{isZero } c_0 &=_{\beta} \text{true}, \\ \text{isZero } c_{n+1} &=_{\beta} \text{false}. \end{aligned}$$

Church Encoding

Predecessor

A predecessor combinator is defined by

$$\text{pred} := \lambda n f x. n (\lambda g h. h (g f)) (\lambda u. x) (\lambda u. u),$$

where

$$\text{pred } c_0 =_{\beta} c_0,$$

$$\text{pred } c_{n+1} =_{\beta} c_n.$$

Church Encoding

Recursion

The factorial function is an example of recursive functions on natural numbers following the schemata

$$\begin{aligned}f &: \mathbb{N} \rightarrow A \\f\ 0 &= a \\f\ (S\ n) &= g\ n\ (f\ n)\end{aligned}$$

where A is a set, $a \in A$, $g: \mathbb{N} \rightarrow A \rightarrow A$ and S is the successor function.

Remark

A reader familiarised with recursion theory will identify the primitive recursion schemata.

Church Encoding

A recursor for natural numbers

Let A be a set. From the previous schemata for (primitive) recursive functions, we define a recursor.[†]

$$\begin{aligned}\text{rec} & : (\mathbb{N} \rightarrow A \rightarrow A) \rightarrow A \rightarrow \mathbb{N} \rightarrow A \\ \text{rec } f \ a \ 0 & = a \\ \text{rec } f \ a \ (\text{S } n) & = f \ n \ (\text{rec } f \ a \ n)\end{aligned}$$

[†]A higher-order function which allow define recursive functions.

Church Encoding

Recursor

Since `rec` is also a recursive function, we can define a recursor for numerals using `fix`.

$$\begin{aligned}h &:= \lambda r f a n. (\text{isZero } n) a (f (\text{pred } n) (r f a (\text{pred } n))), \\ \text{rec} &:= \text{fix } h,\end{aligned}$$

where

$$\begin{aligned}\text{rec } f a c_0 &=_{\beta} a, \\ \text{rec } f a c_{n+1} &=_{\beta} f c_n (\text{rec } f a c_n).\end{aligned}$$

Church Encoding

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Question

What is it necessary for defining `rec`? A fixed-point combinator, (implicit) ‘Boolean’ case analysis, a test for zero and a predecessor function.

Church Encoding

Example

A combinator for the factorial function

$$\text{fac} : \mathbb{N} \rightarrow \mathbb{N}$$

$$\text{fac } 0 = 1$$

$$\text{fac } (S\ n) = S\ n \times \text{fac } n$$

is defined by

$$\text{fac} := \text{rec } (\lambda x\ y. \text{mult } (\text{succ } x)\ y)\ c_1.$$

Church Encoding

Definition

A **number-theoretic function** is a function whose signature is

$$\mathbb{N}^k \rightarrow \mathbb{N}, \text{ with } k \in \mathbb{N}.$$

Church Encoding

Definition

Let φ be a **partial** number-theoretic function $\varphi: \mathbb{N}^k \rightarrow \mathbb{N}$. The function φ is **λ -definable** **respect to the Church encoding** iff there exists a λ -term F such that for all $n_1, \dots, n_k \in \mathbb{N}$,

$$\begin{aligned}\varphi(n_1, \dots, n_k) = a &\Rightarrow F\ c_{n_1} \dots c_{n_k} =_{\beta} c_a, \\ \varphi(n_1, \dots, n_k) \text{ does not exist} &\Rightarrow F\ c_{n_1} \dots c_{n_k} \text{ has no } \beta\text{-nf.}\end{aligned}$$

[†]See, e.g., (Barendregt [1981] 2004, Corollary 6.4.6) and (Hindley and Seldin 2008, Theorem 4.23).

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Theorem

The Turing-computable functions are λ -definable respect to the Church encoding.[†]

[†]See, e.g., (Barendregt [1981] 2004, Corollary 6.4.6) and (Hindley and Seldin 2008, Theorem 4.23).

Numeral Systems

Definition

A **numeral system** is a sequence of combinators

$$d = d_0, d_1, \dots$$

and combinators true_d , false_d , succ_d and isZero_d such that for all $n \in \mathbb{N}$ (Barendregt [1981] 2004, Definition 6.4.1),

$$\begin{aligned}\text{succ}_d d_n &=_{\beta} d_{n+1}, \\ \text{isZero}_d d_0 &=_{\beta} \text{true}_d, \\ \text{isZero}_d d_{n+1} &=_{\beta} \text{false}_d.\end{aligned}$$

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Example

The Church numerals $c = c_0, c_1, \dots$ and the combinators true , false , succ and isZero are a numeral system.

Numeral Systems

Notation

We shall denote by $\mathbf{d}$ a numeral system $d_0, d_1, \dots, \text{true}_{\mathbf{d}}, \text{false}_{\mathbf{d}}, \text{succ}_{\mathbf{d}}$ and $\text{isZero}_{\mathbf{d}}$.

Numeral Systems

Notation

We shall denote by $\mathbf{d}$ a numeral system $\mathbf{d}_0, \mathbf{d}_1, \dots, \mathbf{true}_{\mathbf{d}}, \mathbf{false}_{\mathbf{d}}, \mathbf{succ}_{\mathbf{d}}$ and $\mathbf{isZero}_{\mathbf{d}}$.

Lambda definable functions on a numeral system

Let $\mathbf{d}$ be a numeral system and let φ be a partial number-theoretic function $\varphi: \mathbb{N}^k \rightarrow \mathbb{N}$. The function φ is **λ -definable respect to the numeral system $\mathbf{d}$** iff there exists a λ -term F such that for all $n_1, \dots, n_k \in \mathbb{N}$,

$$\begin{aligned}\varphi(n_1, \dots, n_k) = a &\Rightarrow F \mathbf{d}_{n_1} \dots \mathbf{d}_{n_k} =_{\beta} \mathbf{d}_a, \\ \varphi(n_1, \dots, n_k) \text{ does not exist} &\Rightarrow F \mathbf{d}_{n_1} \dots \mathbf{d}_{n_k} \text{ has no } \beta\text{-nf.}\end{aligned}$$

Numeral Systems

Definition

A numeral system d is **adequate** iff all the Turing-computable functions are λ -definable with respect to d (Barendregt [1981] 2004, Definition 6.4.2ii).

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Example

The Church numeral system is adequate.

Numeral Systems

Theorem

A numeral system d is adequate iff there exists a combinator pred_d such that for all $n \in \mathbb{N}$ (Barendregt [1981] 2004, Theorem 6.4.3),

$$\text{pred}_d d_{n+1} =_{\beta} d_n.$$

Numeral Systems

Theorem

A numeral system $\mathbf{d}$ is adequate iff there exists a combinator $\text{pred}_{\mathbf{d}}$ such that for all $n \in \mathbb{N}$ (Barendregt [1981] 2004, Theorem 6.4.3),

$$\text{pred}_{\mathbf{d}} \mathbf{d}_{n+1} =_{\beta} \mathbf{d}_n.$$

Remark

There are various adequate numeral systems in the literature. See, e.g., Barendregt ([1981] 2004), Goldberg (2000) and Jansen (2013).

References

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